

CPEPRACDSN1 WORKPLAN

(1) Course Title: Computer Engineering
Practice and Design 1

(2) Thesis Title: GeoVision: AI-powered
Construction Monitoring
System

(3) Total Duration (in
months): 6 months

(4) Planned Start Date: August 2026

(5) Planned End Date:
February 2027

(6) OBJECTIVES	(7) TARGET ACTIVITIES	(8) TARGET ACCOMPLISHMENTS (quantify, if possible)	Aug	Sep	Oct	Nov-Jan	Feb
Develop a centralized monitoring dashboard that utilizes OpenCV, a ResNet18 model for construction-stage classification, and a YOLOv8 object detection	• Set up development environment and project structure • Implement API endpoints for photo ingestion and storage • Develop database schema and ORM models • Implement user authentication and authorization • Create real-time data processing pipeline	Fully functional backend API with database integration and authentication system	X	X	X	X	

model implemented in PyTorch to analyze construction progress based on captured images. Implement role-based access control (RBAC) to restrict system functions and information according to authorized user roles. Implement WebSocket communication to provide timely dashboard updates regarding captured images, processing status, connectivity status, delay notifications, and other							
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relevant monitoring events.							
Develop a hardware-based image acquisition system consisting of two fixed-position cameras installed at different diagonal viewpoints, capable of capturing construction site images at a predefined time interval. Implement geotagging technology to verify the authenticity of every captured construction image by recording its corresponding geographic location and	• Prepare construction site for data collection • Configure ESP32-CAM devices with geotagging • Deploy devices to capture construction site photos • Collect and organize construction images with metadata • Validate data quality and completeness	Collection of 500+ geotagged construction photos across different stages	X	X	X		

capture time.							
Develop a centralized monitoring dashboard that utilizes OpenCV, a ResNet18 model for construction-stage classification, and a YOLOv8 object detection model implemented in PyTorch to analyze construction progress based on captured images. Design an automated mechanism capable of detecting construction progress, identifying delays or inconsistencies, and generating visual progress reports for	• Research and select ResNet18 architecture for stage classification • Configure YOLOv8 model for object detection • Set up training pipeline with PyTorch • Implement progress calculation aggregator logic • Create model evaluation framework	AI models trained and validated with >85% accuracy on stage classification		X	X	X	

project stakeholders.							
Develop a centralized monitoring dashboard that utilizes OpenCV, a ResNet18 model for construction-stage classification, and a YOLOv8 object detection model implemented in PyTorch to analyze construction progress based on captured images.	• Define construction stage labels and annotation guidelines • Manually annotate collected construction images • Create training/validation dataset splits • Document annotation process and validation rules • Perform quality assurance on annotations	Annotated dataset with 500+ labeled images ready for model training	X	X	X		
Develop a centralized monitoring dashboard that utilizes OpenCV, a ResNet18 model for construction-stage classification, and a YOLOv8 object detection	• Optimize API performance and database queries • Implement caching and async processing • Add comprehensive error handling and logging • Conduct code review and refactoring • Set up CI/CD pipeline for automated testing	Production-ready backend with optimized performance and automated testing			X	X	

model implemented in PyTorch to analyze construction progress based on captured images. Integrate computer vision, geotagging, hybrid local and cloud storage, and centralized dashboard processing into a unified construction monitoring platform.							
Develop a centralized monitoring dashboard that utilizes OpenCV, a ResNet18 model for construction-stage classification, and a YOLOv8 object detection model	• Fine-tune model hyperparameters • Implement model ensemble techniques • Add confidence score thresholding • Create model serving infrastructure • Document model performance metrics and limitations	Optimized AI models deployed with comprehensive documentation			X	X	

implemented in PyTorch to analyze construction progress based on captured images. Design an automated mechanism capable of detecting construction progress, identifying delays or inconsistencies, and generating visual progress reports for project stakeholders.							
Develop a hardware-based image acquisition system consisting of two fixed-position cameras installed at different diagonal	• Select and procure ESP32-CAM and components • Design PCB layout and housing enclosure • Implement firmware for photo capture and geotagging • Set up wireless connectivity and power management • Create assembly and deployment documentation	Functional ESP32-CAM prototype with firmware and housing	X	X			

viewpoints, capable of capturing construction site images at a predefined time interval. Implement geotagging technology to verify the authenticity of every captured construction image by recording its corresponding geographic location and capture time.							
Develop a hardware-based image acquisition system consisting of two fixed-position cameras installed at different diagonal viewpoints, capable of	• Conduct field testing of ESP32-CAM devices • Verify image quality and GPS accuracy • Test battery life and power consumption • Identify and resolve hardware issues • Document testing results and performance metrics	Hardware validated with documented performance metrics and reliability data			X	X	

capturing construction site images at a predefined time interval. Implement geotagging technology to verify the authenticity of every captured construction image by recording its corresponding geographic location and capture time.							
Develop a hardware-based image acquisition system consisting of two fixed-position cameras installed at different diagonal viewpoints, capable of capturing construction	• Optimize power consumption and battery life • Improve enclosure design for weather resistance • Conduct stress testing and reliability analysis • Create production-ready assembly instructions • Prepare hardware for deployment	Production-ready hardware design with optimized performance				X	

site images at a predefined time interval.							
Integrate computer vision, geotagging, hybrid local and cloud storage, and centralized dashboard processing into a unified construction monitoring platform.	• Review and update architecture section with implemented changes • Revise methodology based on actual implementation • Update results section with performance metrics • Refactor literature review for clarity • Proof-read and format according to thesis guidelines	Complete thesis paper draft with all sections revised and updated		X	X	X	
Develop a centralized monitoring dashboard that utilizes OpenCV, a ResNet18 model for construction-stage classification, and a YOLOv8 object detection model implemented in PyTorch to analyze construction progress based	• Prepare presentation slides and demonstration videos • Conduct mock defense sessions • Gather feedback from advisers and peers • Practice presentation delivery • Document defense date and location	Successful prototype defense presentation with documented feedback			X		

on captured images.							
Integrate computer vision, geotagging, hybrid local and cloud storage, and centralized dashboard processing into a unified construction monitoring platform.	- Analyze feedback from prototype defense - Prioritize and plan revisions - Implement suggested improvements to software and hardware - Update documentation with changes - Document all modifications made	Prototype refined based on defense feedback with documented improvements			X	X	
Integrate computer vision, geotagging, hybrid local and cloud storage, and centralized dashboard processing into a unified construction monitoring platform.	- Prepare production deployment checklist - Deploy API to production server - Configure monitoring and logging systems - Deploy AI models to inference server - Prepare user documentation and training materials	System deployed to production with monitoring and support documentation					X
Integrate computer vision, geotagging, hybrid local and cloud storage,	- Conduct comprehensive system integration testing - Perform stress testing and load testing on API - Test AI model accuracy in production environment	Full system tested and validated with comprehensive test report documenting >95% reliability			X	X	

and centralized dashboard processing into a unified construction monitoring platform.	• Verify hardware reliability over extended periods • Create detailed test reports and analysis						
Integrate computer vision, geotagging, hybrid local and cloud storage, and centralized dashboard processing into a unified construction monitoring platform.	• Finalize thesis document and submit for review • Prepare final presentation and demonstration • Rehearse defense presentation multiple times • Address adviser comments and suggestions • Prepare for questions and follow-up discussion	Successful final defense and thesis approval					X